

Datatypes, Operators & Basic Commands

◆ Datatypes:-

➤ String Data Types:

CHAR – 0 TO 255

VARCHAR – 0 TO 65535

TEXT – 0 TO 255

ENUM – 65535

BLOB – 65535

➤ Numeric Data Types:

INT – 255

INTEGER

FLOAT

DOUBLE

BOOL

➤ Data and times Data types:

DATE

DATETIME

TIMESTAMP

TIME

YEAR

◆ Operators:-

➤ Comparison Operators:-

=	$a=b$
!=	$a!=b$
>	$a>b$
<	$a<b$
>=	$a>=b$
<=	$a<=b$
!<	$a!<b$
!>	$a!>b$

➤ Arithmetic Operators:-

+	$a+b$
-	$a-b$
*	$a*b$
/	a/b
%	$a\%b$

➤ Logical Operators:-

ALL	compare a value to all values
AND	allows the existence of multiple conditions
ANY	compare the value in list according to the condition
BETWEEN	search for values, that are within a set of values
IN	compare a value to that specified list value
NOT	reverse the meaning of any logical operator
OR	combine multiple conditions
EXISTS	search for the presence of a row in a specified table
LIKE	compare a value to similar values using wildcard operator

Database

◆ CREATE DATABASE:-

➤ The SQL **CREATE DATABASE** Statement is used to create a new SQL database.

➤ Syntax:-

CREATE DATABASE database_name;

Eg. **CREATE DATABASE** mydb;

◆ DROP DATABASE:-

➤ The SQL **DROP DATABASE** statement is used to drop an existing database in SQL schema.

➤ Syntax:-

DROP DATABASE DatabaseName;

Eg. **DROP DATABASE** mydb;

◆ RENAME DATABASE:-

➤ SQL **RENAME DATABASE** is used when you need to change the name of your database.

➤ Syntax:-

RENAME DATABASE OldDatabaseName **TO**
NewDatabaseName;

Eg. **RENAME DATABASE** mydb **TO** mydb1;

◆ **SELECT DATABASE:-**

- When you have multiple databases in your SQL Schema, then before starting your operation, you would need to select a database.
- Syntax:-

USE Databasename;
Eg. **USE mydb;**

Table

◆ **What is Table:-**

- Table is a collection of data, organized in terms of rows and columns.
- In DBMS term, table is known as relation and row as tuple.
- A table has a specified number of columns, but can have any number of rows.
- A table is also considered as a convenient representation of relations.

◆ CREATE TABLE:-

- SQL **CREATE TABLE** statement is used to create table in a database.
 - If you want to create a table, you should name the table and define its column and each column's data type.
 - Syntax:-

CRETAE TABLE table_name(column_name1 datatype, Column_name2 datatype...);
- Eg. **CREATE TABLE** student(id int(10), name varchar(20));

◆ **DROP TABLE:-**

➤ A SQL **DROP TABLE** statement is used to delete a table definition and all data from a table.

➤ Syntax:-

DROP TABLE table_name;

Eg. **DROP TABLE** student;

◆ **DELETE TABLE:-**

➤ The **DELETE** statement is used to delete rows from a table. If you want to remove a specific row from a table you should use **WHERE** condition.

➤ Syntax:-

DELETE FROM table_name;;

Eg. **DELETE FROM** student;

DELETE FROM table_name [**WHERE** condition];

Eg. **DELETE FROM** student **WHERE** id=1;

◆ **RENAME TABLE:-**

➤ SQL **RENAME TABLE** is used to change the name of a table.

➤ Syntax:-

alter table old_table_name **rename to** new_table_name;

Eg. **Alter table** student **rename to** student_new;

◆ **TRUNCATE TABLE:-**

➤ A truncate SQL statement is used to remove all rows (complete data) from a table.

➤ Syntax:-

TRUNCATE TABLE table_name;

Eg. **TRUNCATE TABLE** student;